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Mathematics: analysis and approaches
Higher level
Paper 1

8 May 2023

Zone A afternoon | **Zone B** morning | **Zone C** afternoon

Candidate session number

2 hours

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Instructions to candidates

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are not permitted access to any calculator for this paper.
- Section A: answer all questions. Answers must be written within the answer boxes provided.
- Section B: answer all questions in the answer booklet provided. Fill in your session number on the front of the answer booklet, and attach it to this examination paper and your cover sheet using the tag provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[110 marks]**.



Please **do not** write on this page.

Answers written on this page
will not be marked.



Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 7]

The function f is defined by $f(x) = \frac{7x+7}{2x-4}$ for $x \in \mathbb{R}$, $x \neq 2$.

(a) Find the zero of $f(x)$. [2]

(b) For the graph of $y = f(x)$, write down the equation of

(i) the vertical asymptote;

(ii) the horizontal asymptote. [2]

(c) Find $f^{-1}(x)$, the inverse function of $f(x)$. [3]

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2. [Maximum mark: 6]

On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called *The Dragon*. This information is summarized in the following frequency table.

Number of times on <i>The Dragon</i>	Frequency
0	6
1	16
2	13
3	2
4	3

It can be assumed that this sample is representative of all visitors to the park for the following day.

(a) For the following day, Tuesday, estimate

(i) the probability that a randomly selected visitor will ride *The Dragon*;

(ii) the expected number of times a visitor will ride *The Dragon*. [4]

It is known that 1000 visitors will attend the amusement park on Tuesday. *The Dragon* can carry a maximum of 10 people each time it runs.

(b) Estimate the minimum number of times *The Dragon* must run to satisfy demand. [2]

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3. [Maximum mark: 6]

Solve $\cos 2x = \sin x$, where $-\pi \leq x \leq \pi$.

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4. [Maximum mark: 6]

Find the range of possible values of k such that $e^{2x} + \ln k = 3e^x$ has at least one real solution.

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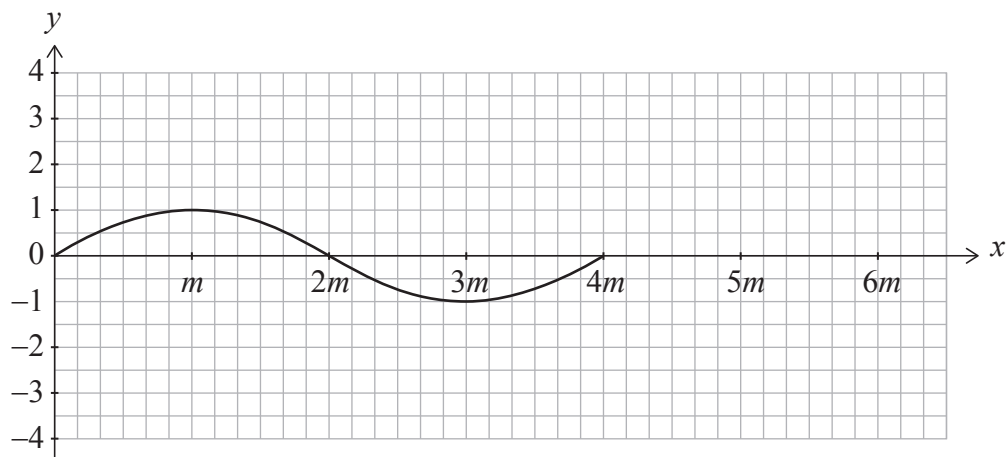
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5. [Maximum mark: 6]

The function f is defined by $f(x) = \sin qx$, where $q > 0$. The following diagram shows part of the graph of f for $0 \leq x \leq 4m$, where x is in radians. There are x -intercepts at $x = 0, 2m$ and $4m$.



- (a) Find an expression for m in terms of q . [2]

The function g is defined by $g(x) = 3\sin\frac{2qx}{3}$, for $0 \leq x \leq 6m$.

- (b) On the axes above, sketch the graph of g . [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

6. [Maximum mark: 5]

The side lengths, x cm, of an equilateral triangle are increasing at a rate of 4 cm s^{-1} .

Find the rate at which the area of the triangle, $A \text{ cm}^2$, is increasing when the side lengths are $5\sqrt{3}$ cm.

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7. [Maximum mark: 6]

Consider $P(z) = 4m - mz + \frac{36}{m}z^2 - z^3$, where $z \in \mathbb{C}$ and $m \in \mathbb{R}^+$.

Given that $z - 3i$ is a factor of $P(z)$, find the roots of $P(z) = 0$.

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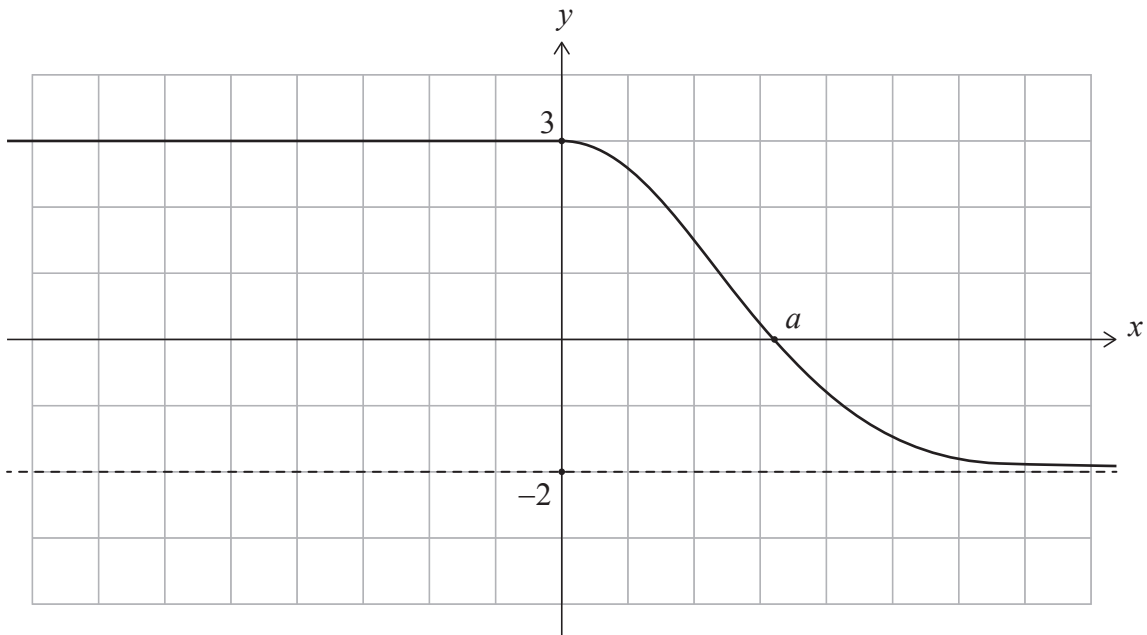
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8. [Maximum mark: 7]

Part of the graph of a function, f , is shown in the following diagram. The graph of $y = f(x)$ has a y -intercept at $(0, 3)$, an x -intercept at $(a, 0)$ and a horizontal asymptote $y = -2$.



(This question continues on the following page)

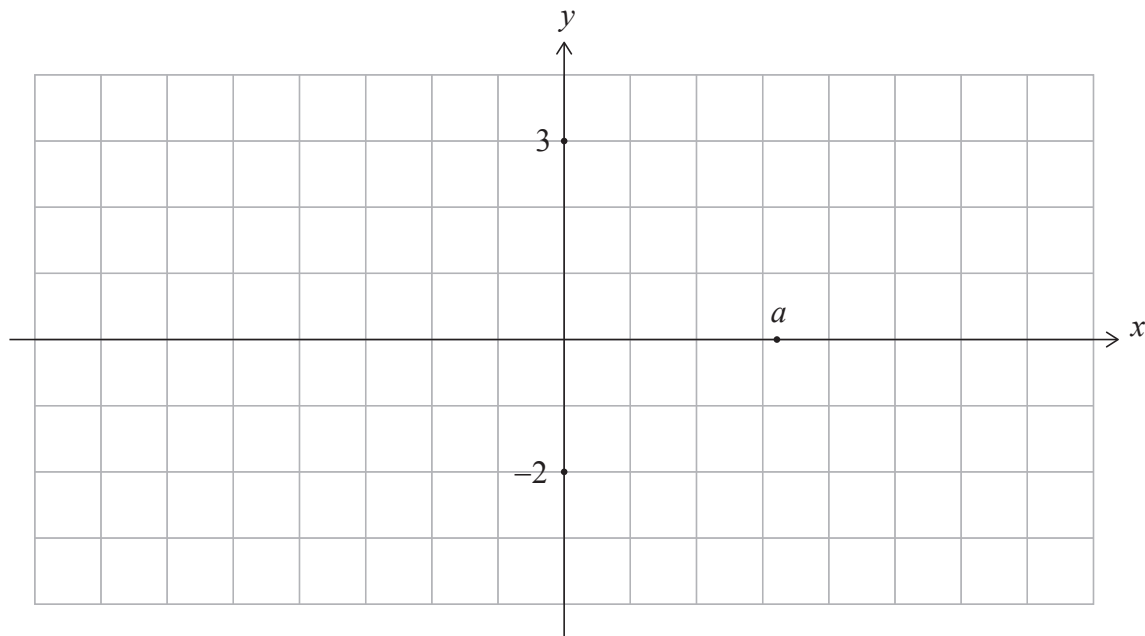


(Question 8 continued)

Consider the function $g(x) = |f(|x|)|$.

- (a) On the following grid, sketch the graph of $y = g(x)$, labelling any axis intercepts and giving the equation of the asymptote.

[4]



- (b) Find the possible values of k such that $(g(x))^2 = k$ has exactly two solutions.

[3]

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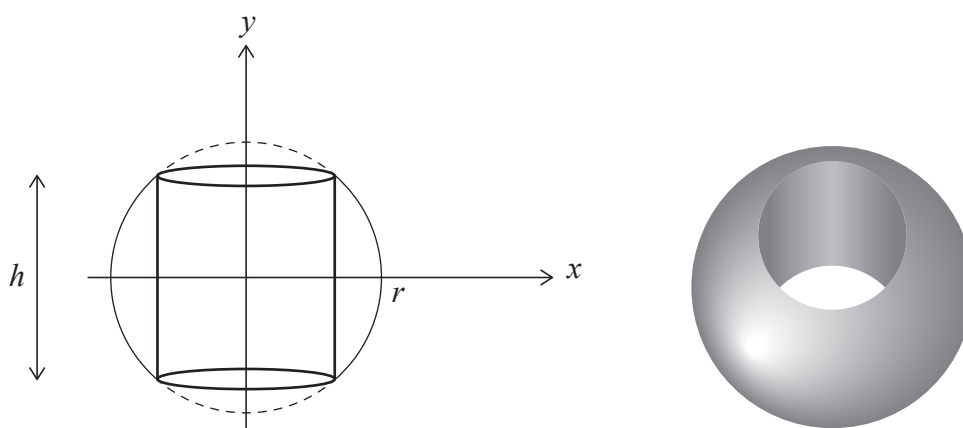
9. [Maximum mark: 7]

The function f is defined by $f(y) = \sqrt{r^2 - y^2}$ for $-r \leq y \leq r$.

The region enclosed by the graph of $x = f(y)$ and the y -axis is rotated by 360° about the y -axis to form a solid sphere. The sphere is drilled through along the y -axis, creating a cylindrical hole. The resulting spherical ring has height, h .

This information is shown in the following diagrams.

diagram not to scale



The spherical ring has a volume of π cubic units. Find the value of h .

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Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 14]

Consider the arithmetic sequence $u_1, u_2, u_3, \dots$

The sum of the first n terms of this sequence is given by $S_n = n^2 + 4n$.

- (a) (i) Find the sum of the first five terms. [4]
- (ii) Given that $S_6 = 60$, find u_6 . [4]
- (b) Find u_1 . [2]
- (c) Hence or otherwise, write an expression for u_n in terms of n . [3]

Consider a geometric sequence, v_n , where $v_2 = u_1$ and $v_4 = u_6$.

- (d) Find the possible values of the common ratio, r . [3]
- (e) Given that $v_{99} < 0$, find v_5 . [2]

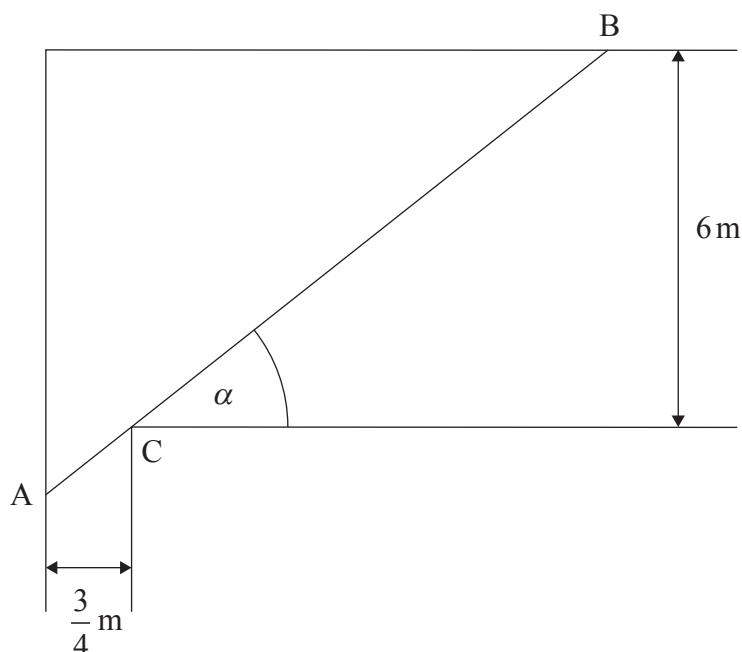


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11. [Maximum mark: 19]

Consider the following diagram, which shows the plan of part of a house.

diagram not to scale



A narrow passageway with width $\frac{3}{4}$ m is perpendicular to a room of width 6 m. There is a corner at point C. Points A and B are variable points on the base of the walls such that A, C and B lie on a straight line.

Let L denote the length AB in metres.

Let α be the angle that [AB] makes with the room wall, where $0 < \alpha < \frac{\pi}{2}$.

(a) Show that $L = \frac{3}{4}\sec\alpha + 6\operatorname{cosec}\alpha$. [2]

(b) (i) Find $\frac{dL}{d\alpha}$.

(ii) When $\frac{dL}{d\alpha} = 0$, show that $\alpha = \arctan 2$. [5]

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(Question 11 continued)

- (c) (i) Find $\frac{d^2L}{d\alpha^2}$.
- (ii) When $\alpha = \arctan 2$, show that $\frac{d^2L}{d\alpha^2} = \frac{45}{4}\sqrt{5}$. [7]
- (d) (i) Hence, justify that L is a minimum when $\alpha = \arctan 2$.
- (ii) Determine this minimum value of L . [3]

Two people need to carry a pole of length 11.25 m from the passageway into the room.
It must be carried horizontally.

- (e) Determine whether this is possible, giving a reason for your answer. [2]

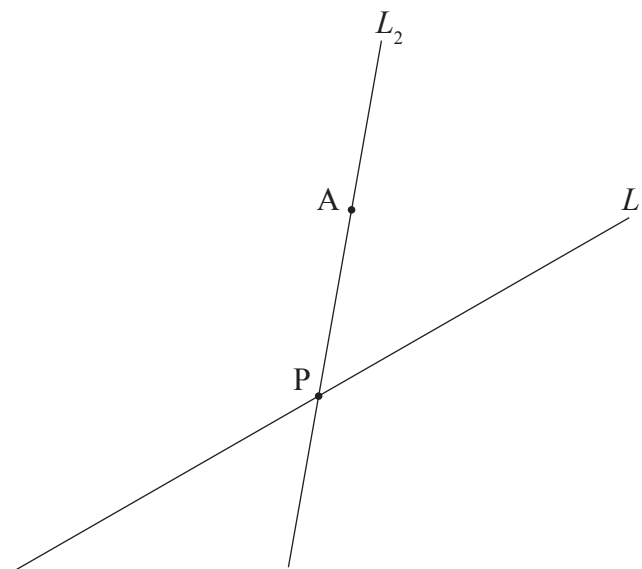


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12. [Maximum mark: 21]

Two lines, L_1 and L_2 , intersect at point P. Point A($2t$, 8, 3), where $t > 0$, lies on L_2 . This is shown in the following diagram.

diagram not to scale



The acute angle between the two lines is $\frac{\pi}{3}$.

The direction vector of L_1 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and $\vec{PA} = \begin{pmatrix} 2t \\ 0 \\ 3+t \end{pmatrix}$.

- Show that $4t = \sqrt{10t^2 + 12t + 18}$. [4]
- Find the value of t . [4]
- Hence or otherwise, find the shortest distance from A to L_1 . [4]

A plane, Π , contains L_1 and L_2 .

- Find a normal vector to Π . [2]

The base of a right cone lies in Π , centred at A such that L_1 is a tangent to its base. The volume of the cone is $90\pi\sqrt{3}$ cubic units.

- Find the two possible positions of the vertex of the cone. [7]

References:

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